

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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Anonymous submission

ABSTRACT

This paper asks how physical scale and population size shape the emergence of complex behavior in open-ended ecological environments, and makes three contributions toward answering it. First, we introduce a new JAX-based multi-agent simulator that pushes objective-free ecological simulation to scales not previously attempted, sustaining populations of over 60,000 agents that each carry their own evolved neural network policy. Crucially, agents in this setting are unsupervised: they receive no explicit rewards or learning objectives, and instead evolve through reproduction, mutation, and selection while continually reshaping their environment and the population around them. Second, using this platform we identify a set of complex behaviors—long-range resource extraction, vision-based foraging, and predation—that emerge purely from competitive and survival pressures rather than from any optimized objective. Third, and most centrally, we show that these behaviors are governed by scale: several of them appear only once the environment and population are sufficiently large, and larger scales make the behaviors that do emerge markedly more stable and consistent. Taken together, these contributions demonstrate that, against a rich history of evolutionary research, scaling ecological simulation on modern hardware opens a promising new direction: treating ecology itself as an instrument of machine learning in an era of abundant compute and efficient frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been studied across a wide variety of biological disciplines, including evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Models of these emergent dynamics range from theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) to the mathematical analyses of population genetics (Fisher, 1930; Wright, 1931).

In these fields, decades of mathematical modeling, laboratory experiments, and field work have tremendously improved our understanding of the mechanisms of evolution in nature. Studying emergent behavior in the natural world, however, faces two persistent obstacles. First, as ecosystems become larger and more complex, they are more difficult to control and measure. Second, even where measurement is possible, running controlled experiments in large-scale natural settings risks damaging or displacing wild populations. Simulation offers a way around both obstacles—but only if it can reach the scales at which interesting ecological behavior actually appears.

This paper takes up that challenge. We study the open-ended evolution of neural network policies in large-scale ecological simulations, where agents have no specified objective or reward signal but instead collect resources to survive and reproduce according to the dynamics of the environment, and where policy changes occur only via mutation as parents pass their policies on to their children. Our central aim is to characterize, at scales not previously possible, how environmental scale and population size shape the emergence of complex behavior in populations of these simple digital organisms.

To make this possible, we build a new JAX-based simulation environment that allows rapid evaluation of large grid worlds: our larger experiments contain over 60,000 individual agents within a

physical area of over 1,000,000 grid cells. Agents must navigate to find food and other resources in order to survive, and can reproduce by collecting enough resources to make a mutated copy of themselves. A 3D visualization of this environment can be found in Figure 1.

We frame this work not only as a way to better understand ecological phenomena through simulation, but also as a way to study ecology as a mechanism for producing machine intelligence. Recent years have shown that in AI, scale doesn't merely improve models—it fundamentally transforms their capabilities (Wei et al., 2022a). Yet many attempts at embodied learning remain confined to relatively small environments with low populations. We therefore ask what capabilities emerge when we scale not just a single model but an entire ecosystem. Just as reasoning emerges in individual models only beyond a certain size threshold (Wei et al., 2022b), we ask which strategies evolve within populations only when environmental size and complexity reach sufficient richness.

Contributions. In answering this question, we make three contributions:

- **A simulator for large-scale ecology.** We introduce a JAX-based, GPU-accelerated multi-agent simulator that supports open-ended populations of over 60,000 evolving agents—orders of magnitude beyond comparable ecological simulations—each running its own neural network policy without any external objective or reward.
- **Emergent behaviors from ecological pressure alone.** Using this simulator, we identify complex behaviors that emerge purely from competition and survival, including long-range resource gathering (“mining”), vision-based foraging, and predation.
- **Scale as a driver of emergence.** We show, through extensive controlled experiments, that both sensing modality and environmental scale strongly shape which behaviors emerge. Several behaviors appear reliably only at large scales, and larger scales make emergent behaviors more consistent and more stable.

Concretely, we show that agents with simple compass sensors can adapt to conduct long-range resource gathering expeditions, but that this behavior emerges less reliably at small scales; and that agents with vision sensors evolve better foraging and attacking behavior than agents without them, with these effects again appearing more reliably at large scales. Together, these results demonstrate that the scale and complexity enabled by modern hardware accelerators offer a promising new approach to studying how evolution in ecological environments can produce rich new behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field's inception, and our contributions extend this lineage to a new scale. Conway's Game of Life (Games, 1970) simulates populations through simple local rules. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown, 1992) simulate life as self-replicating programs that evolve through mutations. Like Conway, Reynolds (1987) used local rules of motion to produce emergent flocking and schooling behaviors.

Researchers have used these principles to evolve the locomotion morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods have evolved neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our first point of departure from these approaches is our objective: rather than producing agents that better adapt to their environment via some reward or objective signal, we seek to understand what behaviors emerge naturally from ecological pressure alone, and how environmental scale and complexity shape those behaviors.

Evolution has also provided inspiration for the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) apply the principles of evolution and natural selection to optimization. Other similar works use evolutionary methods to encourage novelty within the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). Other works have used these approaches to optimize neural networks for tasks like Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018).

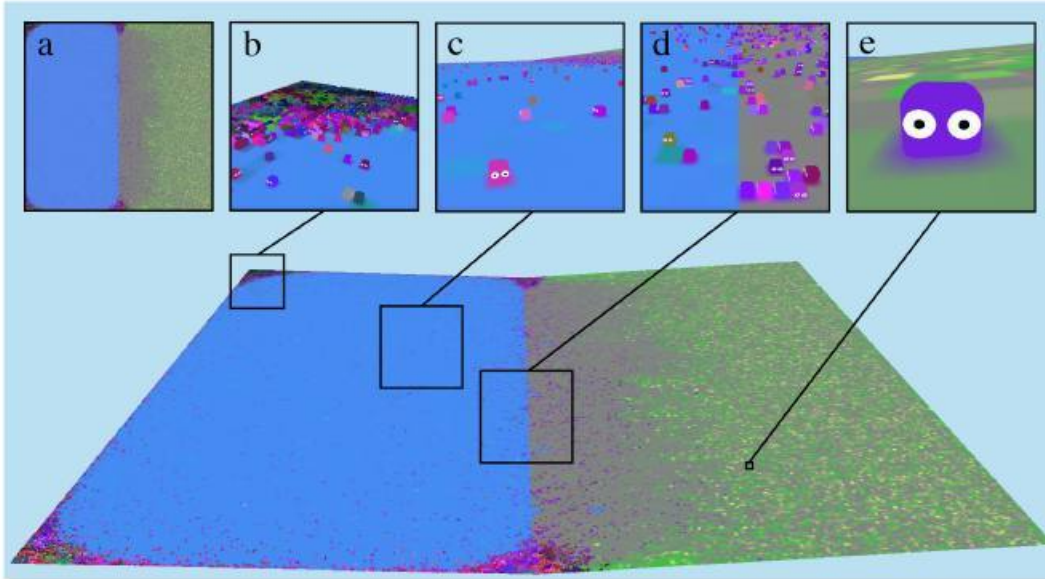


Figure 1: A 3D visualization of a 512×512 map in our simulator, the platform underlying all three of our contributions. Insert (a) shows a top-down view of the entire map, (b) a group of agents clustered in the corner, (c) agents in the water, (d) agents travelling back and forth from the land, and (e) a close-up of a single agent gathering resources.

Here too our contribution is framed differently: while we employ evolutionary mechanisms to update the agents, our focus is on studying emergent behavior rather than developing a novel optimization algorithm or applying an existing one to a fixed objective.

A rich history of work also takes biological inspiration to develop organisms within an environment. Early examples include [Bedau \(1992\)](#), [Yaeger et al. \(1994\)](#), [Channon & Dampier \(1998\)](#), and [Tisue et al. \(2004\)](#). Some recent works use cellular automata to grow organisms with emergent morphologies and behaviors ([Mordvintsev et al. 2020](#); [Robinson, 2021](#); [Heinemann 2021](#); [Randazzo & Mordvintsev 2023](#); [Caussan 2025](#)). [Mordatch & Abbeel \(2018\)](#) and [Park et al. \(2023\)](#) use simulated environments to investigate the emergence of language and communication between agents. The emergence of behavior due to autocurricula ([Leibo et al. 2019](#))—the idea that populations of agents provide an increasingly challenging learning landscape for each other—has also been explored, but in environments with smaller populations than ours ([Leibo et al. 2018](#); [Team et al. 2021](#)). In a complementary line of work, [Wang et al. \(2019\)](#) and [Aki et al. \(2024\)](#) take the reverse approach, evolving environments specifically optimized for ideal agent learning. Though these works evolve the environment rather than the agent, they directly motivate our third contribution: they highlight how environmental complexity—and, we argue, environmental scale—can elicit sophisticated behaviors.

Most closely related to our work are several efforts that simulate populations of agents interacting within complex environments; comparing against them makes clear what each of our contributions adds. [Baker et al. \(2019\)](#) examine a Hide-and-Seek task where a small number of hider agents learn to use environmental tools to evade seeker agents, both trained via reinforcement learning. Other works similarly train populations for specific tasks: playing video games ([Jaderberg et al. 2019](#)), manipulating objects ([Petrenko et al. 2021](#)), or capturing agents ([Zheng et al. 2017](#); [Yamada et al. 2020](#)). While many of these works explore emergence, they do so using fixed tasks and objectives—in contrast to the objective-free setting central to our second contribution. The Neural MMO framework ([Suarez et al. 2019; 2023](#)) takes a more open-ended approach to reinforcement learning, placing agents within a massively multi-agent game setting and optimizing only for survival rather than fixed objectives. However, the population remains capped at a fixed number of agents (experiments examine up to 128 concurrent agents) without generational inheritance, limiting examination of the population dynamics that our simulator is built to expose. [Hamon et al. \(2023\)](#) aligns more closely with our work through variable population sizes, genetically inspired evolution,

and generational timelines, though its environment is significantly smaller, less feature-rich, and does not directly explore the importance of scale—precisely the question our third contribution targets. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but also operates at a much smaller scale (a maximum of 256 agents) and emphasizes tool use through programmable robots. In short, while all of these works take steps toward examining emergent ecological dynamics, a comprehensive understanding requires populations that can grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can emerge. Our contributions integrate exactly these elements, enabling examination of ecological phenomena inaccessible in prior work.

Finally, we are inspired by recent works that emphasize environmental speed and scale, using optimized libraries to increase the computational speed and number of agents able to interact in game-like environments (Koyamada et al. 2023, Matthews et al. 2024, Lechner et al. 2023). Our first contribution builds directly on this progress in computational efficiency, adopting similar JAX-based acceleration techniques to achieve large scales and fast simulation in the ecological setting.

3 METHOD

This section develops the formal and computational machinery behind our first contribution: a scalable, objective-free ecological simulator. We first define the class of games our environment instantiates, then describe the concrete environment, agent policies, and the GPU-accelerated implementation that makes populations of tens of thousands of evolving agents tractable.

3.1 ECOLOGICAL GAMES

We model the interactions between populations of agents and their environment as a type of multiplayer game we refer to as an **Ecological Game (EG)**. Similar to mathematical models such as Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG contains underlying Markovian transition dynamics but is designed to express changing populations of agents in an objective-free setting. Like an SG, an EG contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function T mapping states s_t and a set of actions for each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Unlike an SG, it removes the reward function, and adds a population function $\Gamma(s_t)$ that provides information about which agents in I are currently alive and the identity of their parents so that birth and death events can be inferred. Just as MDPs and SGs have partially observable counterparts, POMDPs and POSGs, an EG can also be partially observable, in which case we refer to it as a POEG. The difference between a POEG and an EG is that a POEG does not allow each agent to observe the entire state, and instead incorporates an observation function $O(s_t, i)$ that maps a state and player to an observation for that player. Our environment is a POEG with different observation functions depending on various sensor configurations explained in the next subsection.

3.2 ENVIRONMENT

The particular POEG used in our experiments is a large grid world containing various resources that agents must collect in order to reproduce and survive. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ where s_t^m is 2D data associated with the map, such as the terrain and distribution of resources, and s_t^a is data associated with the agent, such as its position, health, and internal resources.

Terrain: Each map has a rock layer defining the height of each grid cell. Maps also include water, which is initially added to grid cells with a rock value below a preset sea-level. Throughout an episode, water flows downhill using a discrete flow simulation based on the height defined by this rock layer. Our experiments use six global terrain configurations, shown in Figure 2: **Ocean, Beach, Island, Lake, Isthmus, and Channel**. To support the scale study that is central to our third contribution, we run each at 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grid sizes, scaling the initial population size proportionally to the change in grid area. Once the simulation begins, the population then fluctuates according to the dynamics of the environment.

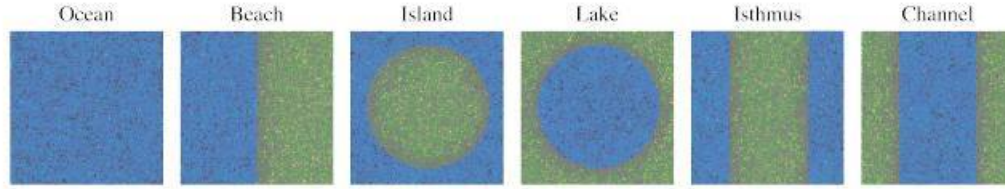


Figure 2: The six terrains used in our experiments, shown as overhead images at 256×256 grid size. Small dark spots are individual agents; green and yellow are concentrations of energy and biomass. Each image was captured 5000 steps into an episode—after most biomass in the water has been eaten, but before agents have evolved to survive on land for longer periods.

Resources and Agent Health: Agents must collect **water**, **energy**, and **biomass** to survive and reproduce in these environments. Initially, biomass and energy are distributed randomly throughout the environment but can be consumed and redistributed by individual agents.

Each action an agent takes expends a small amount of energy and water. The spent water returns to the landscape, where it flows downhill to the nearest body of water, while the energy is used up and destroyed. Biomass is used for reproduction; all agents require a fixed amount of biomass to exist and therefore cannot reproduce until they have collected enough excess biomass to donate to their offspring. Like water, the total biomass remains constant in the environment throughout the simulation, but unlike water, the biomass does not flow downhill or move on its own. Energy is not held constant but slowly grows on free-standing biomass. We designed these resource dynamics to provide agents with a slightly more challenging problem than collecting a single food source. For ablations on resource rules, see Appendix F.2

Agents also have health points (HP) that they must maintain in order to avoid death. If an agent overspends a critical resource, their health will rapidly deteriorate, but they may use energy and water to slowly recover HP over time. Agents may also attack other agents, reducing the targets' HP. They also take a small but increasing amount of damage based on their age, meaning that agents eventually die of old age if not by other causes. When an agent's HP goes to zero and they die, whatever resources they were carrying are returned to the environment in the grid cell they were standing on. In this way, if an agent attacks and kills another agent, it may then consume their resources as a basic form of predation. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Because sensing modality is one of the two levers whose effect our experiments isolate, we give agents four distinct sensors. The **internal** sensor reports the agent's age, health, and carried resources; the **external** sensor detects resources in the agent's own grid cell; the **compass**, inspired by the magnetoreceptors found in birds and other animals, reports the agent's global direction as a 4-way one-hot vector; and **vision** provides a 7×7 top-down rendered patch around the agent, with three color channels plus a fourth channel encoding the local difference in elevation. For ablations on vision configurations, see Appendix F.6

Actions: Agents have a finite set of discrete actions grouped into five categories. The first is **Rest**, which does nothing but costs the least energy. There are three **Move** actions: rotate clockwise, rotate counterclockwise, and move forward one grid cell. Multiple agents cannot occupy the same cell; if an agent tries to move into the same cell as another agent, or if two agents try to move to the same cell, their move action is canceled. The agents also have a single **Attack** action which kills all agents in the 3×3 square directly in front of them. For ablations on attack strength, see Appendix F.7. There are three separate **Eat** actions, one for each of the resource types. If an agent is in the same cell as a particular resource, taking the corresponding eat action will transfer a fixed amount of that resource from the environment to the agent's stomach, which has a fixed maximum capacity.

Finally, agents have a single **Reproduce** action that creates a new offspring via single-parent reproduction. This action is only successful if the agent has accumulated enough resources to donate to its new child. The newly created agent is created directly behind its parent. The child's policy contains a mutated copy of its parent's weights. The child also immediately inherits a fixed amount of its parent's resources.

To turn sensing into a controlled variable, our experiments often disable certain sensors or the attack action, letting us attribute behavioral differences to specific abilities. We use three sensing combinations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**), and append **+A** when attacking is allowed; for example, (**RC+A**) denotes an agent with resource sensors, a compass, and the attack action. These labels recur throughout the experiments as the axis along which our second and third contributions are demonstrated.

3.3 POLICIES

To keep the object of study as simple as possible—so that any emergent behavior we later report is clearly attributable to ecological pressure and scale rather than to policy complexity—our agent policies are implemented as small, memoryless MLPs. Omitting memory simplifies the computational model and reduces network parameters. The raw values for each sensor are flattened, normalized to the range $[-1, 1]$, and fed through separate linear layers with an output dimension of 64. These values are summed and fed through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. The action space is discrete, so the final output is a linear layer with an output dimension equal to the number of possible actions. Actions are sampled from a softmax distribution with a temperature that is allowed to evolve with the network weights. Weights are represented using half-precision bfloat16 values to support larger population sizes and faster run times. Depending on the number of sensors an agent is using, the network size for each agent is roughly between 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5

In all runs, the weights for the initial population of agents use Kaiming initialization (He et al., 2015), and bias vectors are initialized to zero. Policies for agents born mid-episode are created by copying their parent’s weights and applying Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values, a mutation approach similar to that used in other neuroevolution research (Such et al., 2017). While more complex approaches such as crossover are possible, we again opt for the simpler mechanism deliberately: keeping reproduction minimal isolates the effect of environmental scale, which is the central question of our third contribution. Empirically, we found that a larger mutation rate was necessary due to the use of low-precision floating-point numbers, and we verify robustness to this choice with a mutation-rate ablation in Appendix F.4

Agents also have a 3-channel trait controlling their color when rendered onto the map. The color trait mutates similarly to the network weights and determines what an agent looks like when observed by other agents with the vision sensor. In theory, this allows agents to distinguish between different types of other agents.

3.4 COMPUTATION AND REPRODUCIBILITY

The scale that defines our first contribution rests on an efficient implementation. All experiments are implemented in JAX (Bradbury et al., 2018) and run on Nvidia H100 and A40 GPUs using our simulation environment, which computes every part of the environment step on-device. Unless otherwise noted, all experiments run for 2M environment steps or until extinction, with per-seed wall-clock times ranging from under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E describes the simulator architecture in more detail, reports additional scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

Our experiments deliver the evidence behind our second and third contributions: that complex behaviors emerge from ecological pressure alone, and that sensing modality and environmental scale govern which behaviors emerge and how reliably. We organize the results around two families of emergent behavior—long-distance resource gathering (§4.1) and visual foraging and predation (§4.2)—and in each case isolate the roles of sensors and scale through controlled comparisons. A summary of the supporting ablations follows in §4.3

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment establishes two claims that directly support our contributions. First, complex behavior emerges without any objective: agents evolve the ability to travel long distances inland to find resources. Second, this emergence is governed by sensing and scale: in environments with clear directional structure, agents with a compass (**RC**) consistently develop far more effective long-range resource gathering than the baseline (**R**) agents without a compass, and this behavior emerges more consistently and effectively at larger environmental scales.

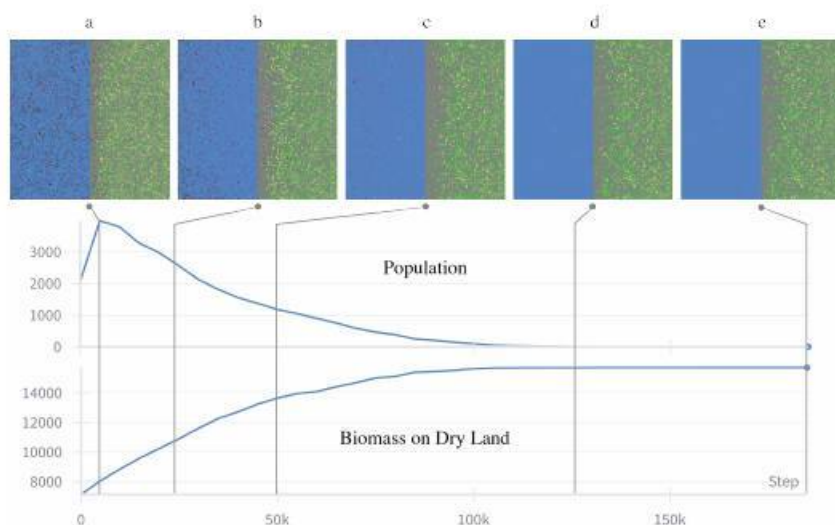


Figure 3: The baseline against which mining emerges: a representative trial with (**R**) agents (no compass) in a 256×256 **Beach** environment. The top chart shows population over time, the bottom chart the free biomass on dry land. Without a compass, biomass accumulates on land and the population collapses.

We begin with a qualitative example of the baseline (**R**) agent class, shown in Figure 3 to illustrate the ecological dynamics against which the compass advantage will stand out. After 5000 time steps (a), all the initial agents on dry land have died due to lack of water. In the meantime, those in the water have rapidly increased in number due to the rich initial resources. At this early stage, the population does not yet contain sophisticated policies that can efficiently seek out resources. From this early peak in population, agents in the water occasionally stumble onto land (b). When they do, they often cannot find their way back to the ocean and die, depositing their biomass on land. By 50,000 time steps (c), this gradual process has nearly doubled the amount of uneaten biomass on land. This transfer of resources from water to land has caused the population to shrink because there is no longer enough biomass in the water to support a large population. By 125,000 steps (d), almost all biomass has been transferred to land, and the population is nearly extinct. The final agent dies around 185,000 steps (e).

This example demonstrates the ecological impact of the agents' inability to find their way back to water: individual deaths do not merely reduce the population, they redistribute resources in ways that reshape the prospects of the entire population. Crucially for what follows, this ecological catastrophe also creates an opportunity—an unexploited niche for any agent that can discover how to walk onto land, forage in the rich deposits of biomass left by its dead ancestors, and safely return to water. The next experiment shows that a compass is what lets agents seize it.

The contrast is stark when we provide agents with an internal compass (**RC**), shown in Figure 4. At first, a similar pattern emerges (a): the population shrinks and free biomass on land rapidly increases. Now, however, some agents use their compass to drift into the relative safety of the corners. At around 150,000 steps (b) some agents begin to “mine” the biomass on land, using their compass to navigate back and forth between land and water. By 200,000 steps (c) these miners have cleared the entire beach. The population then undergoes only minor changes, with agents forming clumps around the edges (d), and remains largely stable for the full 2,000,000 time steps (e). This mining

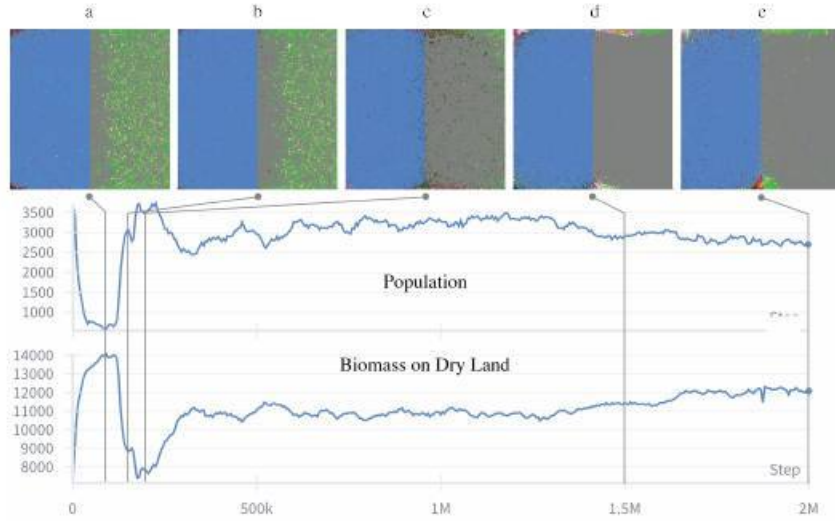


Figure 4: Emergent mining behavior: a representative trial with **(RC)** agents (with compass) in a 256×256 **Beach** environment. The top chart shows population over time, the bottom chart the free biomass on dry land. In contrast to Figure 3, agents evolve to mine the accumulated land biomass, and the population stabilizes for the full run.

behavior—emerging with no objective beyond survival, and unlocked specifically by the compass sensor—is the first concrete instance of our second and third contributions.

Table 1: Evidence for both the sensor and the scale effect: mining events in the **Beach** world for agents with a compass **(RC)** versus without **(R)**, across five world sizes and initial populations. Compass-equipped agents mine far more reliably, and mining becomes more consistent as scale grows (occurring in all seeds at and above 256×256). 🛠 denotes the number of seeds that featured a mining event; ☠ the number of seeds in which the population went extinct before 2M steps. See Table 3 in Appendix A for detailed per-seed data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
	🛠 ☠		🛠 ☠		🛠 ☠		🛠 ☠		🛠 ☠	
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

To move from anecdote to quantitative evidence, we run a full set of **Beach** experiments across grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents respectively. Each experiment uses four random seeds, and we record whether mining behavior emerges—reported as a drop of at least 10% in the free biomass on dry land from the historical high point—along with the onset time of the mining event (the local maximum in biomass on land) and the percentage drop in biomass. Table 1 summarizes mining and extinction events, with full per-seed details in Table 3 in Appendix A. The results confirm both halves of the claim: mining consistently emerges in **(RC)** agents but rarely in **(R)** agents, and it emerges more consistently in larger environments, occurring in all seeds at and above the 256×256 scale. At smaller scales, populations are less stable and more likely to go extinct early.

To put the scale effect on firmer statistical footing, we run additional seeds with **(R)** and **(RC)** agents at 512×512 (32 seeds) and 128×128 (64 seeds) world sizes and use a Kaplan-Meier estimator to measure the probability of extinction and mining as a function of simulator time. We find that extinction is rarer and slower to occur at larger world sizes, while mining is more frequent and happens sooner at smaller grid sizes. A Cox proportional hazards model comparing the timing of

mining discovery and extinction across world sizes and compass usage reinforces the central role of scale: larger world size substantially reduces the extinction rate and enables much more rapid mining discovery than in smaller worlds, and enabling the compass produces an effect of similar magnitude. Full details are in Appendix B

Table 2: The compass advantage generalizes across terrains: mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for agents with a compass (**RC**) versus without (**R**), at the 512×512 grid size. In every terrain, the compass yields more consistent mining and fewer extinctions.  denotes the number of seeds that featured a mining event;  the number of seeds in which the population went extinct before 2M steps. See Table 4 in Appendix A for detailed per-seed data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

To show that the compass advantage is not an artifact of a single terrain, we repeat the experiment across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512×512 grid size with (**R**) and (**RC**) agents. Table 2 reports mining and extinction events, with per-run detail in Table 4 in Appendix A. The pattern holds: equipping agents with a compass yields earlier and more consistent mining as well as more stable populations with lower extinction rates. (**RC**) agents adapt more easily to the island and isthmus than to the lake and channel, where they occasionally go extinct; we attribute this to the fact that walking straight east or west always returns an agent to water in the island and isthmus, whereas no such simple rule exists for the channel and lake. The (**R**) agents cannot reliably adapt to any of these terrains, though they struggle least on the island. Figure 12 in Appendix D visualizes mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

Our second family of experiments extends the same two claims—emergence from ecological pressure, governed by sensing and scale—to a different sensor and two new behaviors. In the **Ocean** environment we ask whether agents can adapt to use vision sensors to locate free resources and to prey on other agents. To isolate the effect of vision, we compare (**RCV+A**) agents (resource sensing, compass, and a 7×7 visual sensor) against (**RC+A**) agents that are identical except for lacking vision.

Figure 5 plots population, biomass utilization, movement actions, and eat actions for agents in the 512×512 and 128×128 grid sizes; plots for other grid sizes appear in Figure 9 in Appendix C with summary statistics in Table 5 in Appendix A. Biomass utilization is the percentage of total environmental biomass currently held inside agents. The vision advantage is clear on every axis: when agents have vision sensors (red lines) they spend less time moving and more time eating, support larger populations, and achieve greater biomass utilization—together indicating that they have adapted to forage more efficiently. The scale effect is equally clear: variance between runs decreases substantially in larger environments. Notably, and somewhat unexpectedly, the vision runs are themselves much more stable, with less variation in these metrics over time.

Vision also reshapes predation, our second emergent behavior in this setting. Figure 6 plots averaged attacks per agent and homicides per attack. Agents with vision attack relatively infrequently but maintain a kill rate of approximately 75%, evidence of specialized, precise predation. In large grids, agents without vision sometimes reach high kill rates as well, but with much more unstable dynamics, and confounding factors make the underlying cause harder to pin down. Visual inspection (Figure 13 in Appendix D) reveals that these high blind-agent kill rates often coincide with clusters of agents building up in one location—clusters that almost never form among visual agents, which may indicate that agents with vision adapt to avoid one another. In smaller environments the gap between visual and blind agents narrows, but populations of visual agents remain far more stable, once again showing scale amplifying the sensor’s benefit.

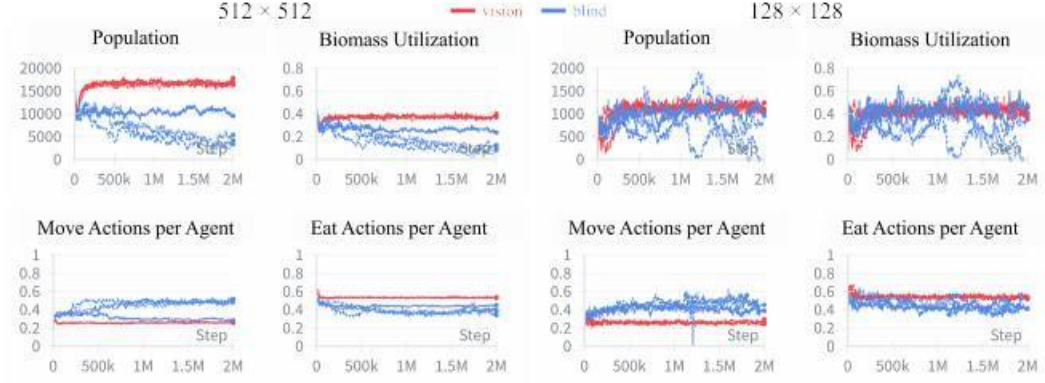


Figure 5: Vision drives more efficient foraging, and scale amplifies the effect. Population, biomass utilization, move actions per agent, and eat actions per agent in 512x512 and 128x128 **Ocean** worlds (the latter three averaged across agents at each time step). Each plot shows four seeds with vision agents (**RCV + A**) (red) and four with blind agents (**RC + A**) (blue). With vision, populations are larger and more stable and biomass utilization is higher, driven by less moving and more frequent, effective eating. At larger scales these dynamics are more stable with less cross-run variance, indicating that vision offers an increasing advantage over blindness as the environment grows. See Figure 9 in Appendix C for grid sizes 1024x1024 and 256x256.

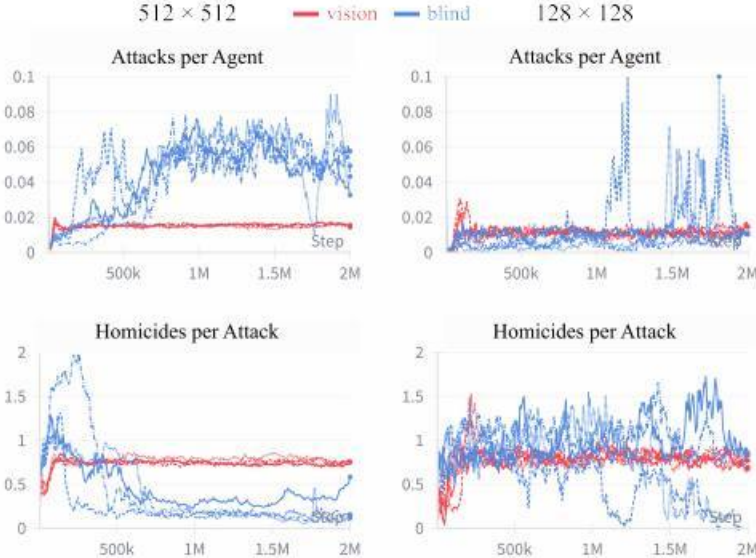


Figure 6: Vision also gives rise to specialized predation. Attacks per agent and homicides per attack, averaged across agents, in 512x512 and 128x128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with much higher precision—the signature of an emergent, targeted predation strategy—and, as with foraging, the effect is more pronounced at larger scales. See Figure 10 in Appendix C for grid sizes 1024x1024 and 256x256.

4.3 ABLATIONS

To confirm that the two contributions above are robust rather than artifacts of a particular configuration, Appendix F reports ablations on initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8), measuring the sensitivity of the emergent behaviors to simulator configuration. Most directly

in support of our scale contribution, Appendix F.3 verifies that the dynamics observed in larger environments are not simply a matter of self-averaging over many small worlds.

5 CONCLUSION

This paper set out to characterize how scale shapes the emergence of complex behavior in open-ended ecological environments, and it delivered on three fronts. We introduced a JAX-based simulator that sustains open-ended populations of over 60,000 evolving agents; we used it to surface complex behaviors—mining, visual foraging, and predation—that arise purely from ecological pressure, with no explicit objective or reward; and we showed through controlled experiments that both environmental scale and sensory richness strongly govern which of these behaviors emerge and how consistently. Taken together, these contributions highlight open-ended ecological settings as a powerful paradigm for investigating the emergence of intelligence, and suggest that ecological and evolutionary pressures, when coupled with sufficiently rich and large environments, may offer a natural pathway toward increasingly intelligent artificial agents. That we can evolve such behaviors in large populations within a reasonable hardware budget, and without any explicit objective, is itself a central takeaway. We hope these results, and the simulator we release, will enable further exploration of ecology as a mechanism for developing and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Per-seed detail behind our sensor and scale contributions: mining events in the **Beach** world for agents with a compass (**RC**) versus without (**R**), across five scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

1024x1024 32768				512x512 8192				256x256 2048				128x128 512				64x64 128				
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x			140	✓	70k	85.3	675
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Per-seed detail showing the compass advantage generalizing across terrains: mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** for **(RC)** versus **(R)** agents. Grid size is 512×512 with an initial population of 8192.

Island					Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Per-seed population and foraging data quantifying the vision advantage for Vision versus Blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, the percentage of total biomass in the system that is currently inside the agents' stomachs. denotes move actions per agent, the fraction of time an agent took a move action, averaged across agents. Similarly, denotes the proportion of agents' actions that took an eat action. Reported values of , , , and are averages of the metric over all steps for the seed.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Per-seed predation data quantifying the vision advantage for Vision versus Blind agents in the **Ocean** world across scales.  denotes attacks per agent, the fraction of time an agent took the attack action, averaged across all agents. The  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps for the seed. Note the outlier of 0.562 for  of seed 3 in the 128×128 (**RC+A**) run. This seed went extinct before 2M steps, resulting in a spike in the attacks per agent metric due to dwindling population size.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

This appendix provides the survival-analysis evidence behind our scale and sensor contributions, comparing rates of extinction and mining emergence as a function of world size and agent access to global direction via the compass sensor. Extinction is defined as the population reaching zero individuals, and mining behavior as in earlier experiments (a 10% drop in free biomass on land).

Runs are grouped based on both world size and compass access. We simulated 64 runs each of 128×128 with no compass and 128×128 with compass, and 32 runs each of 512×512 with no compass and 512×512 with compass. For each of the four groups, we fit a non-parametric Kaplan-Meier estimator to the time-to-extinction distribution (treating runs that did not go extinct within 2M steps as right-censored). In Figures 7 and 8 we observe that worlds tend to survive longer and produce mining more quickly if they are large or if agents have compass information.

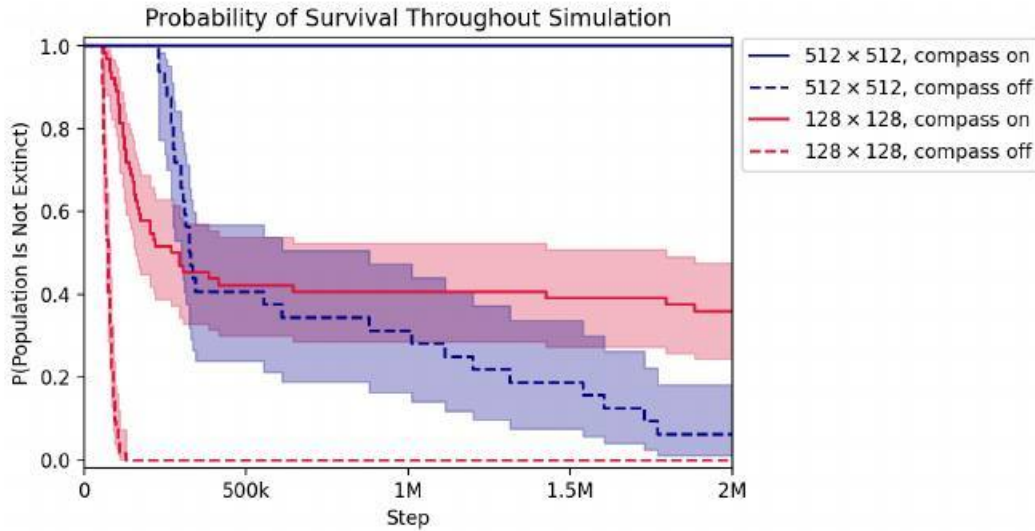


Figure 7: Probability the population has not gone extinct given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan-Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

To attach effect sizes to these two factors, we fit a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7); runs without an emergence or extinction event by 2M steps are treated as censored. Both factors move the outcome in the direction our contributions predict. For extinction, both larger world size and compass use substantially reduce the hazard: the extinction rate in 128×128 is about 25 times that in 512×512 worlds ($HR=0.04$), and enabling the compass produces an effect of similar magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining much more rapidly ($HR=6.07$), and compass use accelerates emergence dramatically ($HR=34.59$). The models discriminate well between groups (concordance 0.80 for extinction, 0.85 for mining). The mechanism, we argue, is that larger worlds sustain larger populations—reducing the chance of hitting the absorbing zero boundary while simultaneously exploring a larger phenotype space through greater total mutation than smaller worlds afford.

C ADDITIONAL FORAGING AND PREDATION RESULTS

These additional scales reinforce the two claims of our vision experiments: that foraging and predation emerge from vision, and that scale sharpens the effect.

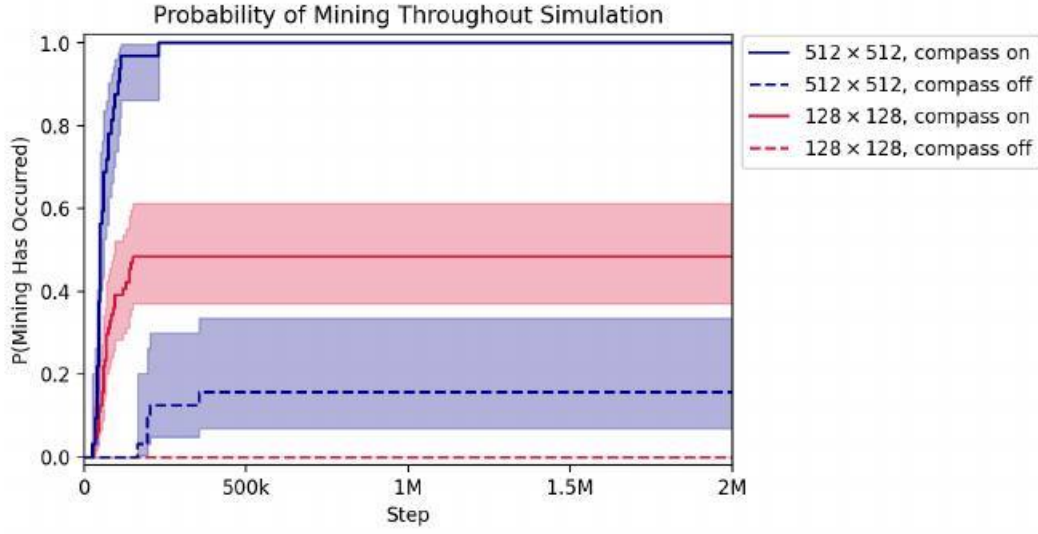


Figure 8: Probability mining has evolved within the population given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan-Meier estimator for each group of runs (e.g. 512×512 , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each 512×512 group fits the estimator on 32 runs; each 128×128 group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events with a 128×128 , compass off baseline. Hazard ratios (HR) less than 1 indicate reduced hazard (longer time to event); for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate of 128×128 . There are 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the 512×512 world, and 64 seeds with compass on and 64 seeds with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

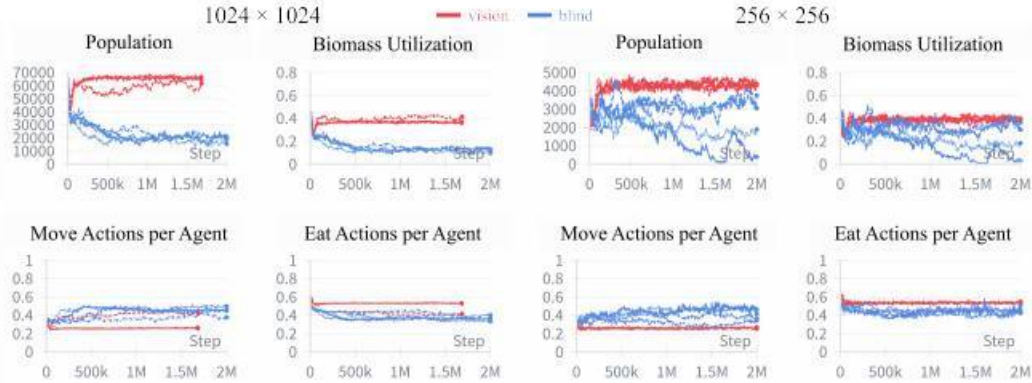


Figure 9: Foraging behaviors in the **Ocean** world at additional scales, confirming the vision advantage reported in the main text. Enabling vision (red) yields more stable populations and greater biomass utilization through more frequent and effective eating and less movement, and these dynamics grow more stable—with less variance across runs—at larger scales. Due to GPU-hour limits, the vision seeds at 1024×1024 ran for 1.7M rather than 2M steps.

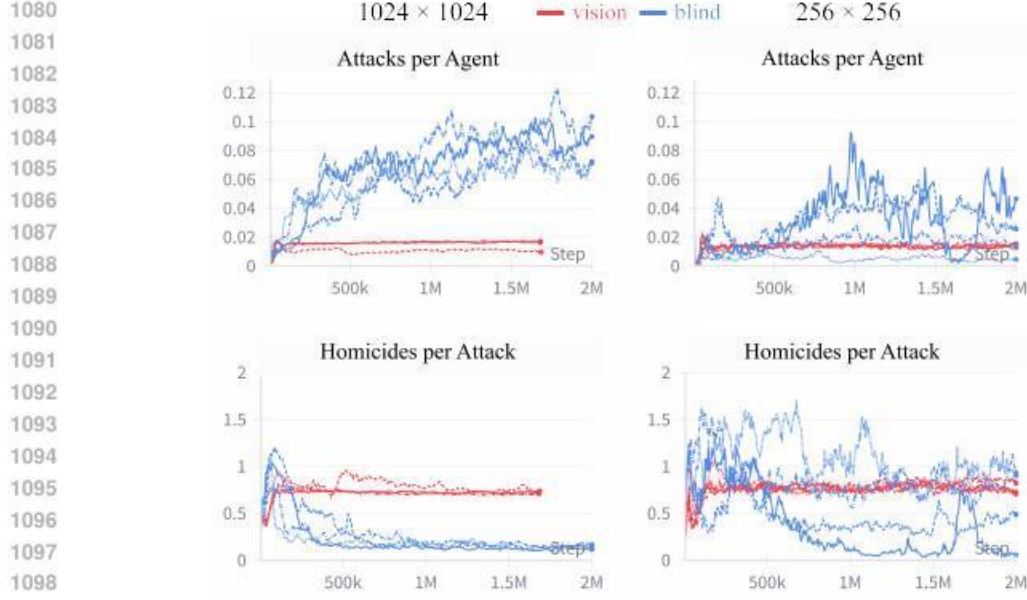


Figure 10: Predation behaviors in the **Ocean** world at additional scales. With vision (red), agents attack less frequently but with much higher precision, evidence of specialized predation emerging alongside the foraging capabilities of Figure 9. As elsewhere, the effect is more pronounced at larger scales, while blind agents (blue) show more variance at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

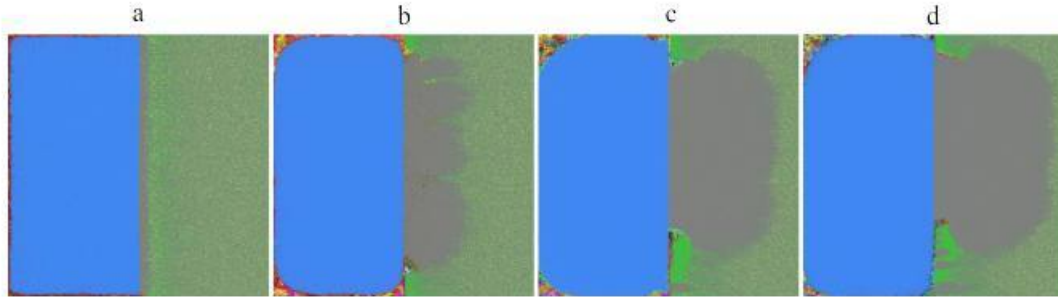


Figure 11: A qualitative illustration of our scale contribution: larger environments induce more varied and complex ecologies with unique spatial patterns. In this 1024×1024 **Beach** world, (RC) agents initially died at the shore (a), transferring biomass to dry land. They then evolved mining behavior, clearing biomass from the beach (b). When biomass re-accumulated at the ends of the beach (c), subpopulations independently re-evolved mining, traveling inland on these mini biomass beaches to extract the resources (d).

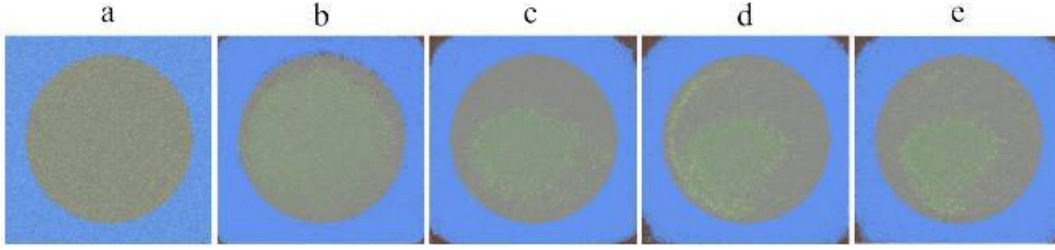


Figure 12: Mining, one of our emergent behaviors, generalizes across terrains. In this 512×512 **Island** world with (RC) agents, different regions of the shore are mined over the course of the run (b), (c). Biomass returns to parts of the island once mining becomes difficult and agents die before reaching water (d), and agents then adapt to re-mine this wall of accumulated biomass, clearing it from the shore (e).

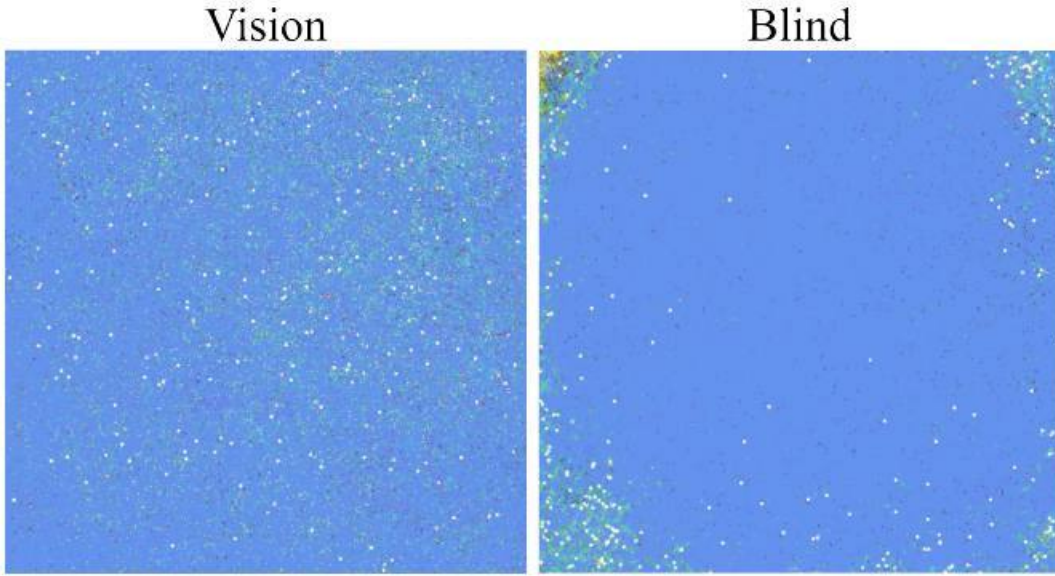


Figure 13: The vision advantage made visible: representative runs in a 512×512 **Ocean** world with attacking enabled—vision (RCV+A) agents on the left, blind (RC+A) agents on the right, with white squares marking an attack. With vision, agents adapt to locate free resources throughout the ocean and prey on others more effectively (see Figures 5 and 6). Blind agents explore for resources far less, instead evolving policies that cluster in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

This appendix expands on our first contribution, the simulator, reporting details on its design, runtime, and experiment reproducibility.

Simulator design. We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Its defining feature is that every part of the environment step—terrain dynamics, resource flows, agent updates, and measurement—is computed on the GPU, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation is deliberately general—it lets the same environment implementation be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules—and is part of what makes the simulator a contribution beyond this paper’s specific experiments. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, en-

abbing fusion into a small number of GPU kernels. This is what yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

Beyond raw throughput, a second component of this contribution is the simulator’s built-in tooling. A measurement API computes and records aggregate statistics—biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters—at each step, feeding directly into our figures and tables. An interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating the very qualitative phenomena our second contribution reports—long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap left by existing systems, which is what makes it a contribution in its own right. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), or Griddly (Bamford et al., 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. Our simulator instead combines all three of GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics—enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper and a reusable foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 quantifies the throughput that makes our scale contribution practical, across grid sizes and maximum agent populations. Each run was conducted on a single H100 GPU for a maximum of 2M steps. The table reports run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total); actual runtime varies with agent network size, available sensors, and logging level.

Even our largest runs remain within a modest budget: at the 1024×1024 grid size they take ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, using roughly 60GB of memory. **Ocean** runs are slightly slower due to the enabled agent attack action and vision sensing.

Table 8: Runtime data in the **Beach** and **Ocean** worlds across different grid sizes and maximum populations of agents. The (RC) agents each have a 22,465 parameter neural network. The (RCV + A) agents each have a 22,529 parameter neural network. Simulation steps per second (hz) and total extrapolated wall-clock runtime in hours for 2M steps are reported. Effective runtime is less for seeds in which populations go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Because our environments are large and randomized, precise outcomes can differ substantially from run to run; we therefore report four random seeds per experiment and, throughout the paper, draw contribution-level conclusions only from patterns that hold across seeds. Due to nondeterminism in some low-level GPU operations, repeating an experiment with the same random seed does not reliably produce identical results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that change rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory; the use of 16-bit weights can coarsen these rounding effects but is not the primary source. Despite this run-to-run variation, the clear high-level patterns that underpin our conclusions emerge robustly once results are aggregated across seeds.

F ABLATIONS

The ablations in this appendix serve our contributions in two ways: they establish that the emergent behaviors we report are robust to specific hyperparameter choices rather than tuned artifacts, and they further probe the central role of environmental scale. We present a series of ablations investigating the effects of specific hyperparameters and the robustness of our results under different environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

A natural worry about our scale contribution is that the effects we attribute to world size are really about initial population size. This ablation separates the two, measuring how initial population size affects mining consistency and population stability in a 256×256 **Beach** world. Results are reported for **(RC)** agent populations; agents without a compass **(R)** had a 0% mining rate and 100% extinction rate across all initial population sizes except the highest tested (8192), which reached 25% mining and 75% extinction. Note that mining and extinction frequencies need not sum to 100%: a population may develop mining and then go extinct, or survive the full 2M steps without ever mining.

The result supports our interpretation. As Figure 14 shows, initial population size has no significant effect on mining consistency or extinction rates: populations starting with only 128 agents evolve mining as frequently as those starting with 8192, and extinction rates are largely stable across initial population size. The scale effect we report is thus a property of the environment’s size, not merely of how many agents it starts with.

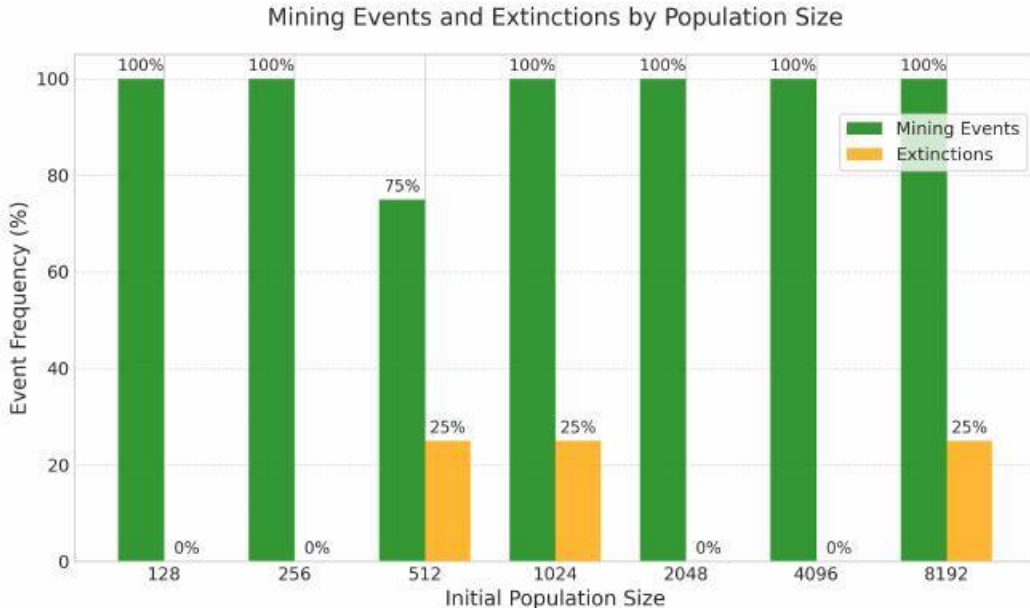


Figure 14: Initial population size does not drive our results: frequency of mining and extinction events in a 256×256 **Beach** world across initial sizes of **(RC)** agent populations. Frequencies are computed over four random seeds per initial population size; e.g. 25% Extinction Event Frequency means the population reached 0 before 2M steps in 1 of 4 seeds.

Figure 15 explains why the effect is so small. Regardless of initial population size, each run rapidly converges to a similar population within the first 50,000 steps. This suggests that the environment implicitly specifies a carrying capacity set by the fixed resources available: populations below it grow to reach it, and populations above it are constrained down to it. The effective population size is therefore similar after an early correction, leaving mining and extinction frequencies unaffected. Larger initial populations do contribute somewhat to total system resources, since agents

carry biomass that is deposited upon death, and so reach slightly higher maximum populations on average (Table 9)—but not enough to shift mining or extinction frequency in this setting. Finally, while population size converges early, it does not stay fixed: later in a run, populations across seeds can diverge substantially as agents evolving different abilities at different times reshape the later resource distribution. This is precisely the kind of long-horizon ecological dynamic our large-scale simulator is built to capture.

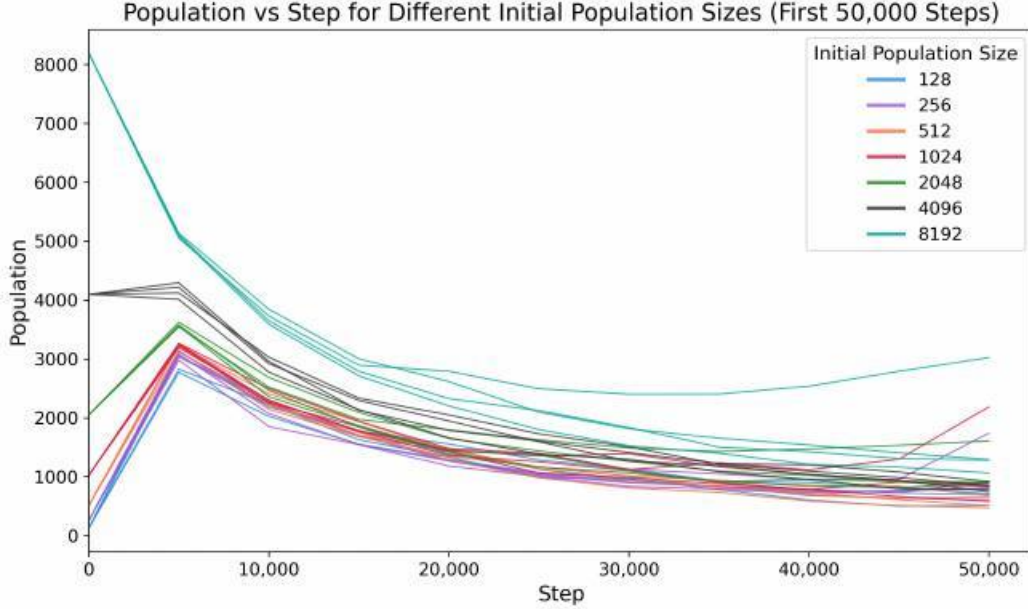


Figure 15: Why initial population size washes out: population during the first 50,000 steps in the 256×256 **Beach** with (RC) agents, four seeds per initial population size. The environment imposes an implicit carrying capacity set by available resources, which all initial population sizes approach early in the run. Larger initial populations add somewhat to total system resources via biomass deposited on death, raising this capacity slightly.

Table 9: The maximum population size reached after step 0 for different starting population sizes, with averages and standard deviations across four random seeds. Larger initial population sizes reach slightly higher maximum populations with greater variance across seeds than for lower initial population sizes. However, the max population size reached is largely robust to the initial population size: an initial population size (128) of only 1.6% of an initial size of 8192 reaches 78.9% of the max population reached by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

F.2 VARYING RESOURCE RULES

Having shown that initial population size does not drive our results, we now identify a factor that does: the resources available in the environment. We evaluate two resource hyperparameters—biomass photosynthesis rate and initial biomass site density—on mining and extinction frequency,

in the same setting as Appendix F.1 with initial population fixed at 2048 and agents again equipped with a compass (RC).

The biomass photosynthesis rate controls the rate at which landscape biomass creates energy via photosynthesis, providing resources for agents over time. The initial biomass site density parameter determines the fraction of grid cells that have biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

The contrast with the previous ablation is telling: whereas initial population size did not affect mining and extinction consistency (Figure 14), changing resource rules has a significant effect (Figures 16 and 17).

The mechanism is the same carrying-capacity argument that underlies our scale contribution. Larger biomass photosynthesis rates or initial biomass site densities directly boost the effective carrying capacity by adding resources to the system, which supports larger populations throughout the run, raising the likelihood of mutations favoring mining while lowering the chance of extinction. Our data bear this out: on average (omitting seeds that went extinct), runs with initial biomass site density 1/16 reached a max population of 1608 after step 0, rising to 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

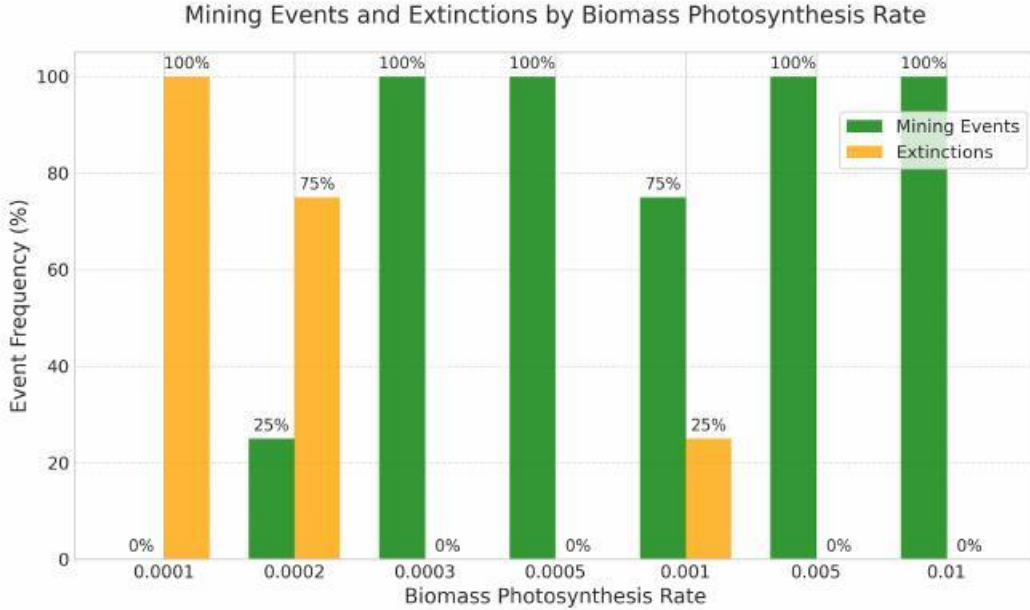


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world for different settings of the biomass photosynthesis rate resource parameter. Frequencies are computed across four random seeds for each setting. Mining consistently emerges with biomass photosynthesis rates of 0.0003 and greater.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

Because scale is the pivot of our third contribution, we must rule out a deflationary explanation for it: that the increased stability of mining behavior in large environments (Table 1) is merely self-averaging rather than a qualitatively different dynamic. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, such that a single large system behaves like the average of many smaller systems of the same total size. Under this view, a 512×512 **Beach** world might simply act like many loosely coupled 128×128 worlds run in parallel, and our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

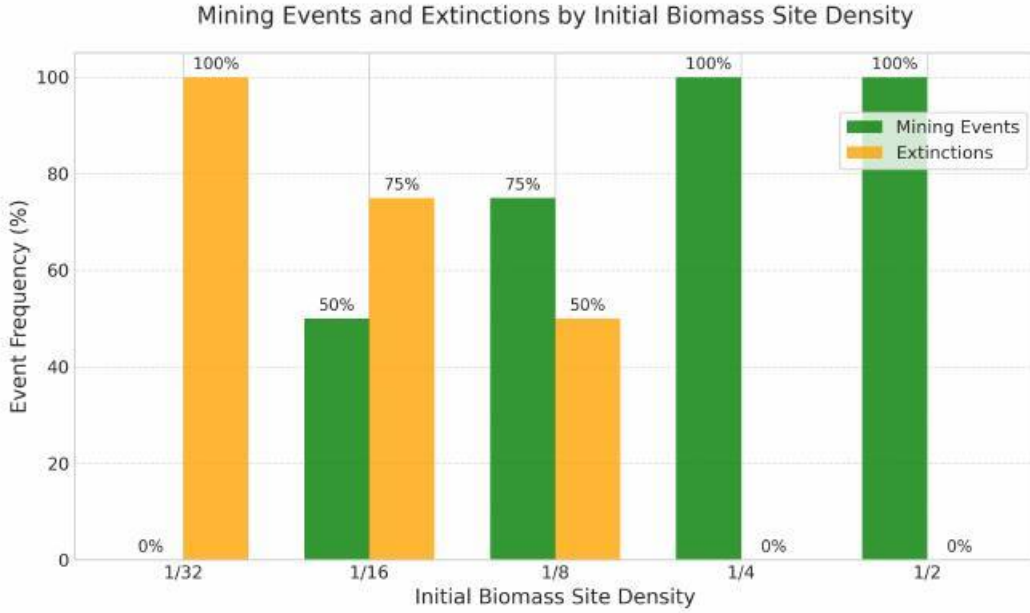


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world for different settings of the initial biomass site density resource parameter. Frequencies are computed across four random seeds for each setting. Greater initial biomass site densities result in more consistent discovery of mining and lower chance of extinction.

To test this hypothesis, we run an experiment holding the total “sample size” roughly fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat this protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments were acting as self-averages of many smaller ones, we would expect mining and extinction rates to be roughly invariant to how we distribute grid cells and agents across runs: 4 large worlds or 64 small ones would exhibit similar probabilities of discovering mining. Instead, we observe a strong dependence on world size. For (**RC**) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

These results are inconsistent with a self-averaging explanation, and thereby strengthen our scale contribution: larger environments genuinely enable and stabilize mining behavior in ways that cannot be reproduced by averaging over more small simulations.

Even when the total number of cells, agents, and simulation steps is matched across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent

Table 10: Mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , 128×128 with initial populations scaled proportionally (8192, 2048, 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (RC) and without a compass (R).

	512×512		256×256		128×128	
	8192		2048		512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

and extinctions are rare, while dispersing the same resources across many small worlds gives populations that frequently fail to discover mining and are more likely to collapse. Increasing environmental scale, then, is not merely smoothing out stochastic noise; it is altering the underlying ecological dynamics—for example by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies.

F.4 MUTATION RATE SWEEP

To show that the mining behavior central to our second contribution is not an artifact of one mutation setting, we test its sensitivity to the mutation rate, which controls the scale of Gaussian noise added to a child agent’s policy at birth and thus how different a child is from its parent. We again run a 256×256 **Beach** world with an initial population of 2048 (RC) agents for up to 2M steps.

Figure 18 shows that mining emerges in at least 3 of 4 seeds for mutation rates between 6×10^{-3} and 7.875×10^{-2} , confirming that the Section 4.1 results at rate 3×10^{-2} are robust across this interval. As expected, rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction, while rates that are too large (e.g., 3×10^{-1}) undermine population stability, consistently causing extinction with no mining.

F.5 NETWORK SIZE

This ablation does double duty: it checks robustness to network architecture and, in doing so, provides another window onto our scale contribution. We study how the size and architecture of the agent policy network affect mining and extinction frequency, and how environmental scale modulates that effect, in 256×256 and 512×512 **Beach** worlds with initial populations of 2048 and 8192 (RC) agents respectively (scaling by four with the number of grid cells). Baselines with (R) agents again yielded rare to no mining and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, yielding 22465 parameters per agent (Table 11). Figure 19 shows that in the 256×256 **Beach**, halving channels to 32 (9185 parameters) or doubling to 128 (61313 parameters) with layers fixed has minimal to no effect on mining and extinction frequency. Network depth, however, matters greatly at this scale: all populations with just 1 layer and 32 channels went extinct, as did 50% of populations with 3 layers and 128 channels.

The key finding directly reinforces our scale contribution: scaling the environment to 512×512 produces markedly different dynamics for networks of varying depth (Figure 20). In the larger world, populations with 3-layer, 128-channel networks consistently mined and survived through 2M steps, and, more surprisingly, so did populations with the small 1-layer, 32-channel networks. We hypothesize that both smaller and larger networks struggle to discover new behaviors, for different reasons: smaller networks have too little capacity to represent good solutions, while larger networks face a search space so large that promising behavior takes longer to find. Whatever the mechanism, the fact that these networks fail so often in small environments yet succeed in larger ones is itself

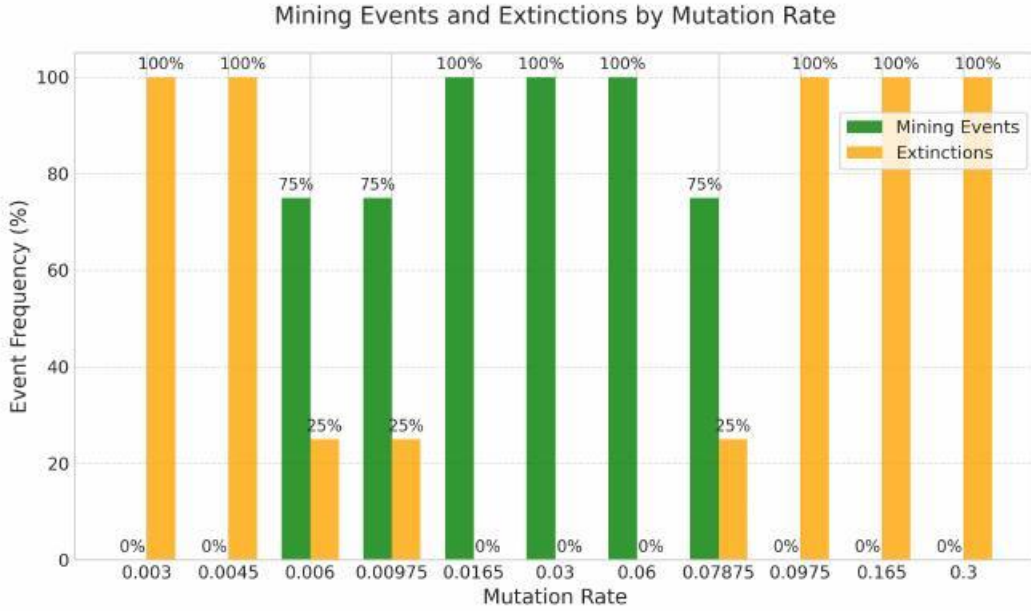


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world for different mutation rates. Frequencies are computed across four random seeds for each setting.

further evidence of the stability that scale confers. A larger environment supports larger populations, allowing more exploration in policy space; simultaneously, locally successful mutants take longer to propagate across a large environment, which preserves diversity and again promotes exploration. For further discussion of these effects, see (Mitchell et al., 2006).

Table 11: Total parameters in (RC) and (R) agent policies with different network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

F.6 VARYING VISION RANGE

Having established that vision drives foraging and predation, we now probe how much vision is needed, sharpening our second contribution. We assess the effect of different vision ranges and viewing configurations on predation and foraging in the 512×512 **Ocean** world with (RCV+A) agents, evaluating visual sensor ranges of 3×3 , 7×7 , and 9×9 . For each range we test two viewing configurations shown in Figure 21 (a) Around, in which agents see cells all around them, and (b) Front, in which agents cannot see cells behind them. The total number of cells in the agent’s vision range is equal for the two configurations. The main vision experiments in Section 4.2 used the 7×7 Around setting.

These results locate the boundaries of the vision benefit. A small 3×3 range yields behavior largely similar to no vision at all (Blind), while 9×9 looks much like 7×7 —evidence of diminishing returns to larger vision ranges in the current configuration. This is intuitive: because agents have no

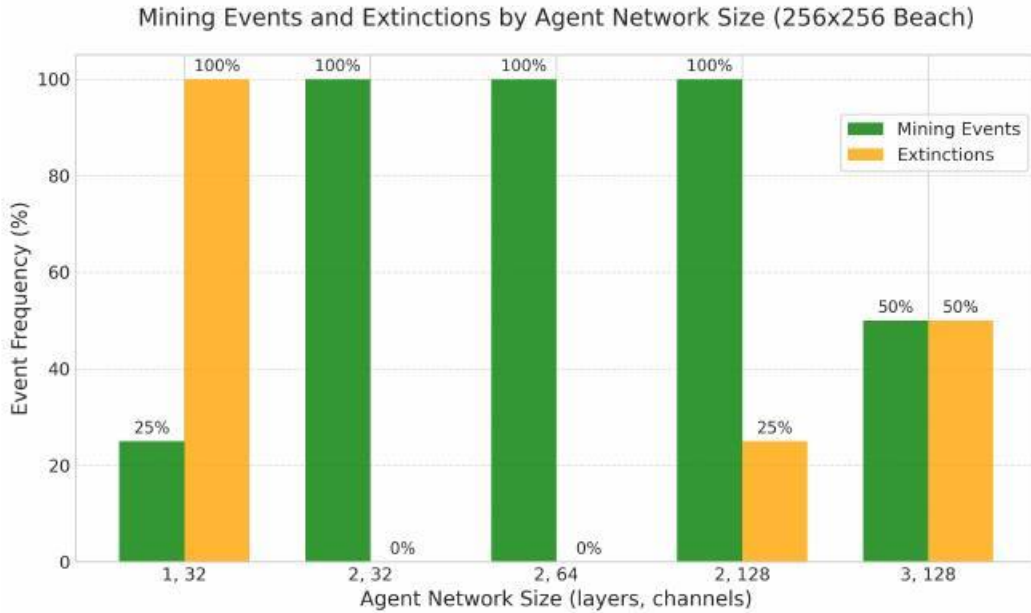


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

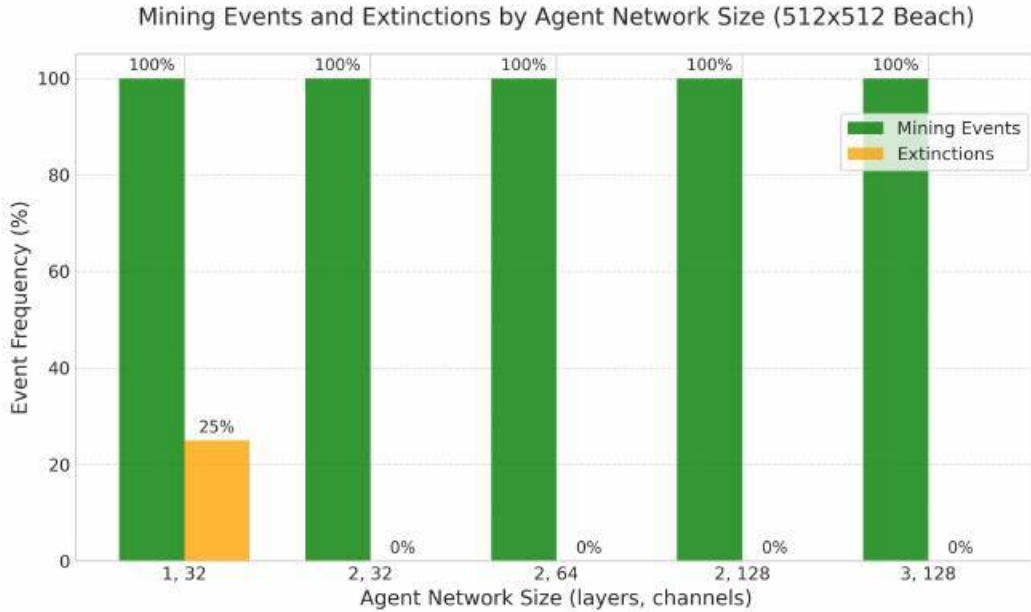


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

memory, visual features at the periphery of a 9×9 window are unlikely to alter the optimal action relative to a 7×7 window often enough to provide a strong learning signal.

The data also speak to viewing geometry. At the 7×7 and 9×9 ranges, the Around configuration is slightly more favorable for population stability and the effectiveness of attack and eat actions, giving higher average population , biomass utilization , and homicides per attack  than the Front configuration. We again attribute this to diminishing returns on distant pixels: in the Around

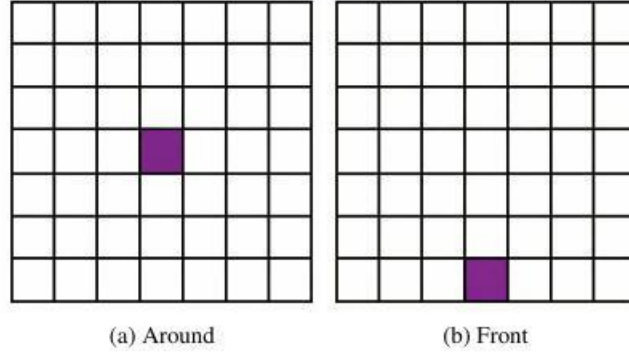


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell represents an agent, facing the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) with varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

configuration, more of the visible cells are nearby and thus more relevant to near-term actions than the distant cells visible under Front. At the 3×3 range the trend reverses, with Front outperforming Around—indicating that here the marginal value of seeing one more pixel ahead outweighs that of seeing one pixel behind, and that 3×3 Around behaves most like the Blind setting.

F.7 VARYING ATTACK STRENGTH

This ablation examines how the predation behavior we report depends on the environment’s attack rules. We vary agent attack strength in the 256×256 **Ocean** world across three settings: (1) Full Attack, dealing 10 HP of damage and instantly killing any agent in the attack radius; (2) Half Attack, dealing 5 HP; and (3) Quarter Attack, dealing 2.5 HP. The main **Ocean** experiments in Section 4.2 used Full Attack.

Reducing attack strength from Full to Half sharply reduces both the frequency and the effectiveness of attacks: attacks per agent and homicides per attack drop considerably (Figure 22). Consequently, lower-attack populations grow larger, with agents moving and eating more and achieving greater biomass utilization. Dropping further to Quarter Attack yields diminishing returns in both reducing violence and enabling further growth. The mechanism is a compounding one: when attacks do less damage, each homicide requires more attacks, and as lethality falls, attacking becomes less useful and is therefore selected for less strongly—so low-attack agents both attack less and become less skilled at it, spending the freed time foraging.

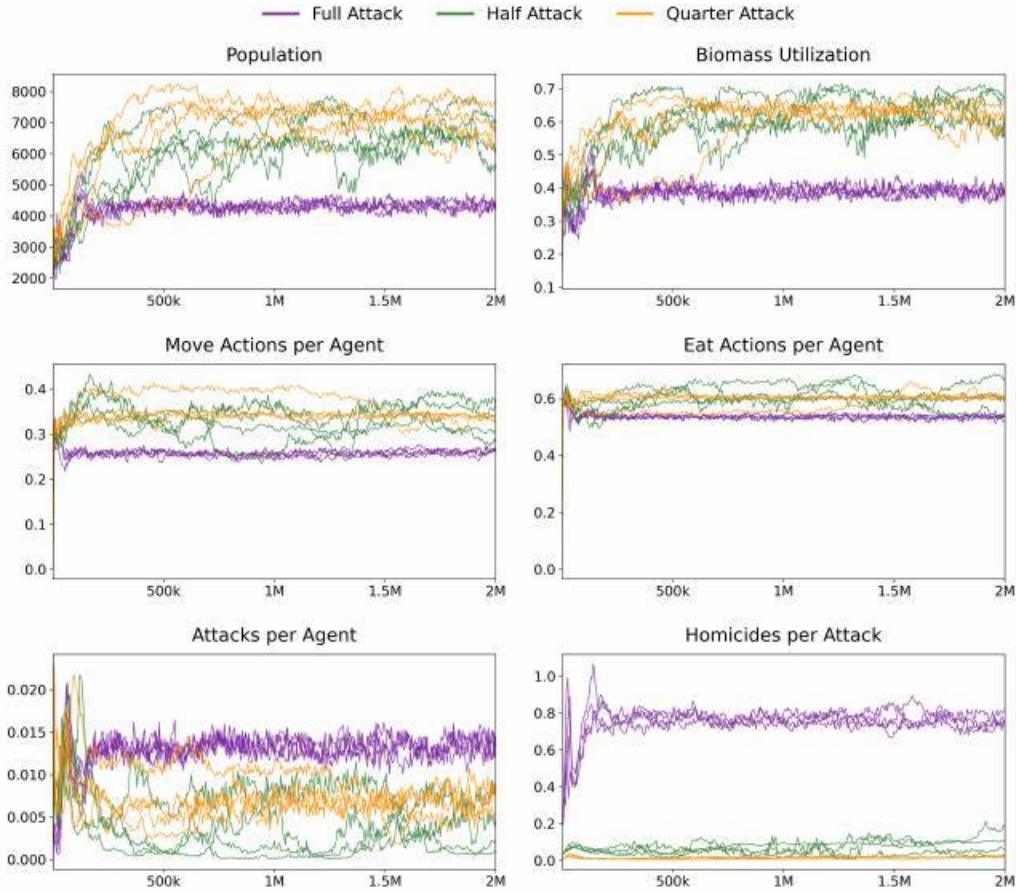


Figure 22: Population and action frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per attack strength setting.

F.8 VARYING AGENT HP

Our final ablation asks whether the emergent behaviors persist when agents face pressure to ‘grow’ over their lifetime. We reduce the starting HP of child agents and lower the healing rate—the amount of HP an agent regenerates per healing step by converting available energy—so that survival demands early resource accumulation. Agents can influence healing only by maintaining enough energy.

We compare two settings: (1) Full HP, with child HP starting at 5 and a healing rate of 1 HP per 0.1 energy; and (2) Low HP, with child HP starting at 1 and a healing rate of 0.5. Max agent HP is 10 in all settings, and the main experiments (Section 4) used Full HP.

The emergent behaviors are largely unchanged: across the two settings there is no significant difference in homicides per attack, eat-action frequency, or biomass utilization (Figure 23). Full HP produces slightly larger populations, while Low HP produces agents that attack less often and move slightly more. We hypothesize that agents born with lower HP must prioritize early-life foraging in order to grow to full size, accounting for this difference.

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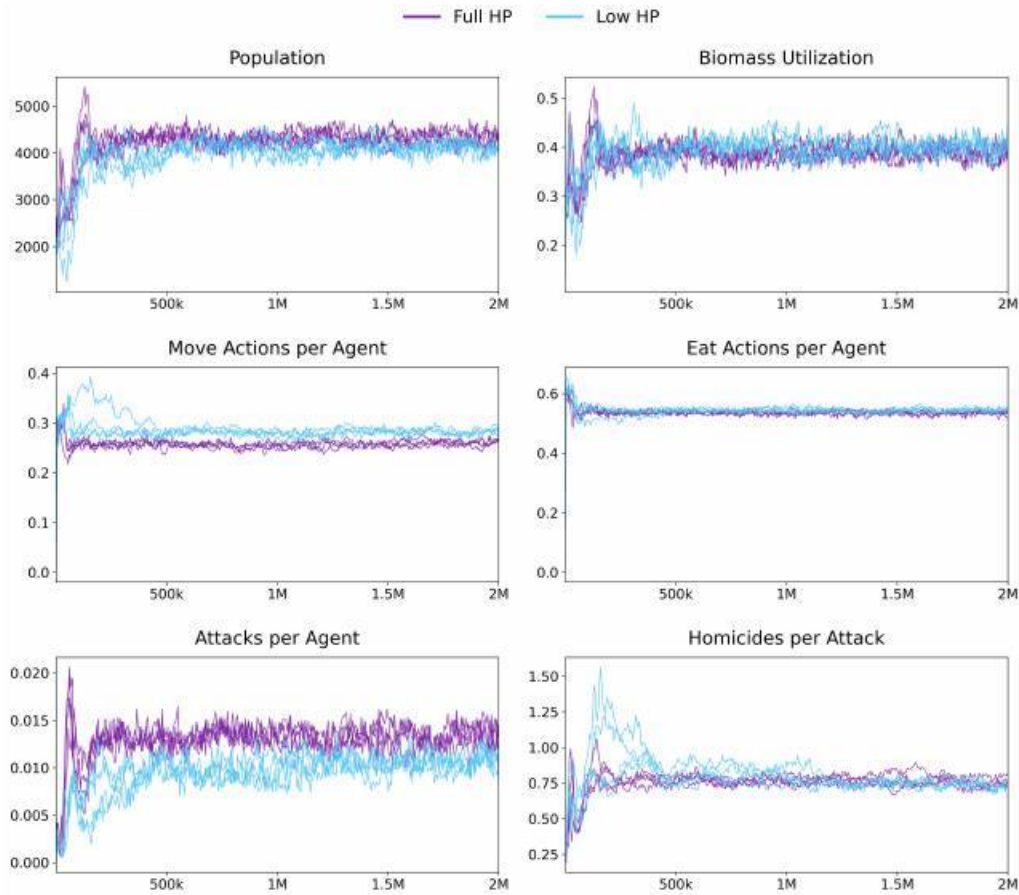


Figure 23: Population and action frequency data over 2M steps in the 256x256 **Ocean** world for **(RCV+A)** agents across Full HP and Low HP settings. Four seeds are plotted per HP setting.